

Drill app guide



WARNING



IGNORING INSTRUCTIONS HAZARD!

To avoid death or injury you **MUST** read, understand and follow operator's and maintenance manuals before installing, inspecting, operating, servicing, testing, cleaning, transporting, storing, dismantling or disposing of the product or a part or accessory of the product. Keep this publication for future reference.

This page is intentionally left blank

Table of Contents

1	Drill app guide.....	7
1.1	The purpose of the application.....	7
1.2	The application PDF files.....	7
1.3	The general adjustment procedure.....	7
1.4	Machine compatibility.....	7
1.5	Challenging ground conditions.....	7
1.6	Hole control.....	7
1.7	Power reduction.....	7
1.8	The rotation calibration.....	7
1.9	The power levels.....	8
1.10	Using intermediate power.....	8

This page is intentionally left blank

1 DRILL APP GUIDE

1.1 The purpose of the application

This application is designed to support the setup of minimum and maximum drilling pressures and rotation speed settings. The optimal parameters depend on ground conditions, flushing, and selected rock tools. The application does not automatically provide exact correct values but guides the operator to a safe and effective operating range.

1.2 The application PDF files

You can close the PDF files by swiping from left to right on the screen.

1.3 The general adjustment procedure

1. Record the current machine parameters.
2. Adjust drilling pressures and rotation speed using the application.
3. Monitor the drilling performance. If the performance improves, keep the adjusted values.

1.4 Machine compatibility

Values are based on i-series drilling rigs. On non-i-series machines, the rotation torque pressure target is typically 15...20 bar higher due to different pressure measurement locations.

1.5 Challenging ground conditions

In soft or fractured ground, it may be difficult to find optimal settings. Lower drilling power often improves performance. Airmist flushing can help in unstable or collapsing holes. A smooth start and proper hole flushing before inserting the second tube may improve drilling performance.

1.6 Hole control

To reduce hole deviation in long-hole drilling, use a guide tube combined with a Retrac bit, and apply controlled and steady pre-collaring and collaring. Airmist flushing can also help.

1.7 Power reduction

If the drilling power is frequently reduced automatically, check and understand the Power Reduction parameters.

1.8 The rotation calibration

The rotation speed should be calibrated 2–3 times per year and always after replacing the rotation proportional valve. The calibration procedure requires the use of a tachometer and reflective tape. All drilling-related proportional valves must be calibrated after replacement.

1.9 The power levels

0% power on the app defines minimum values:

- Minimum rotation torque pressure target
- Minimum percussion pressure
- Minimum feed pressure
- Collaring rotation speed based on the selected bit

100% power represents maximum drilling parameter limits, which must not be exceeded in terms of percussion.

1.10 Using intermediate power

If a certain power level, e.g., 70%, provides good performance, use the app values and set them as new machine maximum parameters:

- Maximum rotation torque pressure target
- Maximum percussion pressure
- Maximum feed pressure
- Drilling rotation speed

The maximum feed pressure values in the application are designed for uphole drilling. Downhole drilling generally requires lower feed pressure.

Note! ***When making adjustments, always use new drill bits to ensure optimal settings.***



www.sandvik.com